Inverted MAML Card 1: Round 1 Thermal Bit-Slicing Report

Hardware: Card 1 (T120+T20, 1k/2.2k) | **FW:** v2.2-MAML (0x20261030) | **Params:** 40% PWM, K=1, LR=0.03, L2=1024, Mom=0.9, Clamp=±75 LSB

1. Round 1 Summary & Cumulative Comparison (R1–R1)

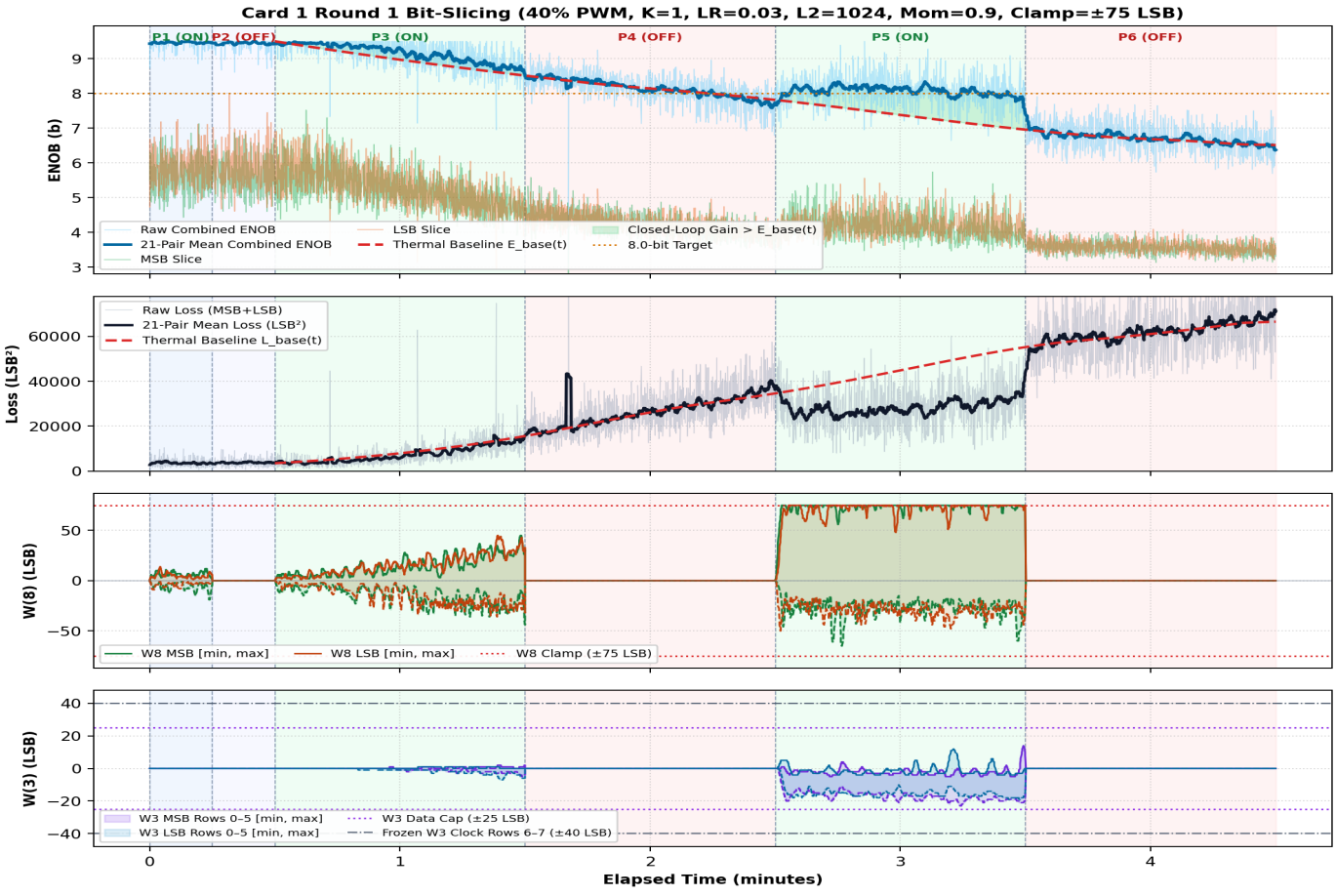
• **Cold Verification:** Phase 1 ON = **9.45 b**, Phase 2 OFF = **9.45 b** (W8=0 strictly).

• **Round 1 Detrended Trend Metrics:** Lift vs E_base = **+0.66 b**, Loss Ratio = **1.57x**, Climb = **+0.74 b**, Jitter = **0.33 b**, Recovery = **0.67**, Score = **+1.957**.

Rnd	LR	L2	Mom	Clamp	Lift	LossR	Climb	Jitt	Recov	Score
R1*	0.03	1024	0.9	±75	+0.66b	1.57x	+0.74b	0.33b	0.67	+1.957

2. Phase-by-Phase Telemetry (4,289 Pairs)

Phase	Regime	N	Start->Late (Mean)	Δ vs E_base	Loss Start->Late	W8 Pk [M,L]	W3 Pk	W8 nz
P1	Cold / Regler ON	312	9.43->9.46 (9.46b)	-	3492->3478	[19,14]	0	311
P2	Cold / Regler OFF	269	9.46->9.44 (9.45b)	-	3129->3695	[0,0]	0	0
P3	Warm / Regler ON	1082	9.44->8.88 (9.17b)	+0.21 b	3970->11035	[45,45]	7	1081
P4	Warm / Regler OFF	973	8.43->7.89 (8.14b)	-0.05 b	17507->32520	[0,0]	0	0
P5	Warm / Regler ON	850	7.74->7.99 (8.07b)	+0.88 b	36416->30612	[75,75]	23	850
P6	Warm / Regler OFF	803	6.96->6.59 (6.71b)	+0.02 b	54265->65928	[0,0]	0	0



3. All W(8) (7x6 MSB+LSB) & W(3) (8x6 MSB+LSB) Weight Dynamics — Round 1

• **Full 84 W(8) + 96 W(3) Weight Coverage:** Displays the signed `[min, max]` excursion envelopes of all 42 `W(8) MSB`, 42 `W(8) LSB`, and 72 `W(3)` active data synapses (capped at ±25 LSB), plus the 24 frozen `W(3)` clock staircase weights in rows 6–7 (`[-40..+40]` & `[+40..-40]` LSB). Full per-synapse 180-trace trajectories are logged natively from Round 4 onward.

Round 1 W(8) & W(3) Bank Coverage (84 W8 + 96 W3 Synapses; Clamp=±75, M3 Cap=±25)

